

Machine Learning Project: Insurance Premium Prediction

Day 1: Comprehensive Data Exploration & Business Analysis | July 28, 2026

1. Executive Summary & Business Context

The insurance industry relies heavily on actuarial science and predictive modeling to accurately assess risk and determine appropriate premium pricing for clients. Mispricing premiums can lead to significant financial losses if set too low, or loss of market competitiveness if set too high. This machine learning project, 'Insurance Premium Prediction', aims to bridge that gap by leveraging historical data and advanced regression techniques to automate and optimize property insurance premium calculations.

This Comprehensive Day 1 Report documents the foundational phase of the machine learning pipeline: Data Ingestion, Exploratory Data Analysis (EDA), and Data Quality Assessment. During this phase, we meticulously unpacked the dataset to understand the underlying statistical properties, business implications of missing records, and the distribution of the target variable ('Total_Annual_Premium_INR'). By combining programmatic analysis with domain understanding, we have laid a robust foundation for downstream feature engineering and model training.

2. Data Ingestion & Structural Overview

2.1 Technical Setup and Environment

The exploratory phase was conducted using an interactive Jupyter Notebook environment, utilizing the Python data science stack. The primary libraries employed include:

- Pandas: For highly optimized, memory-efficient data manipulation and DataFrame structuring.
- NumPy: To facilitate advanced numerical operations and array broadcasts.
- Matplotlib: For generating publication-quality static data visualizations.

2.2 Dataset Dimensions and Feature Space

The raw dataset, 'dataset.csv', was successfully ingested into a Pandas DataFrame. The dataset exhibits a robust dimensionality of 10,000 observations (rows) across 22 distinct features (columns). This volume of data is highly suitable for training stable regression models. The features encompass a wide variety of domain-specific data points, including structural integrity (Property_Age, Construction_Type), external risk factors (Crime_Risk, Natural_Disaster_Risk), and client financial health indicators (Credit_Score, Customer_Income).

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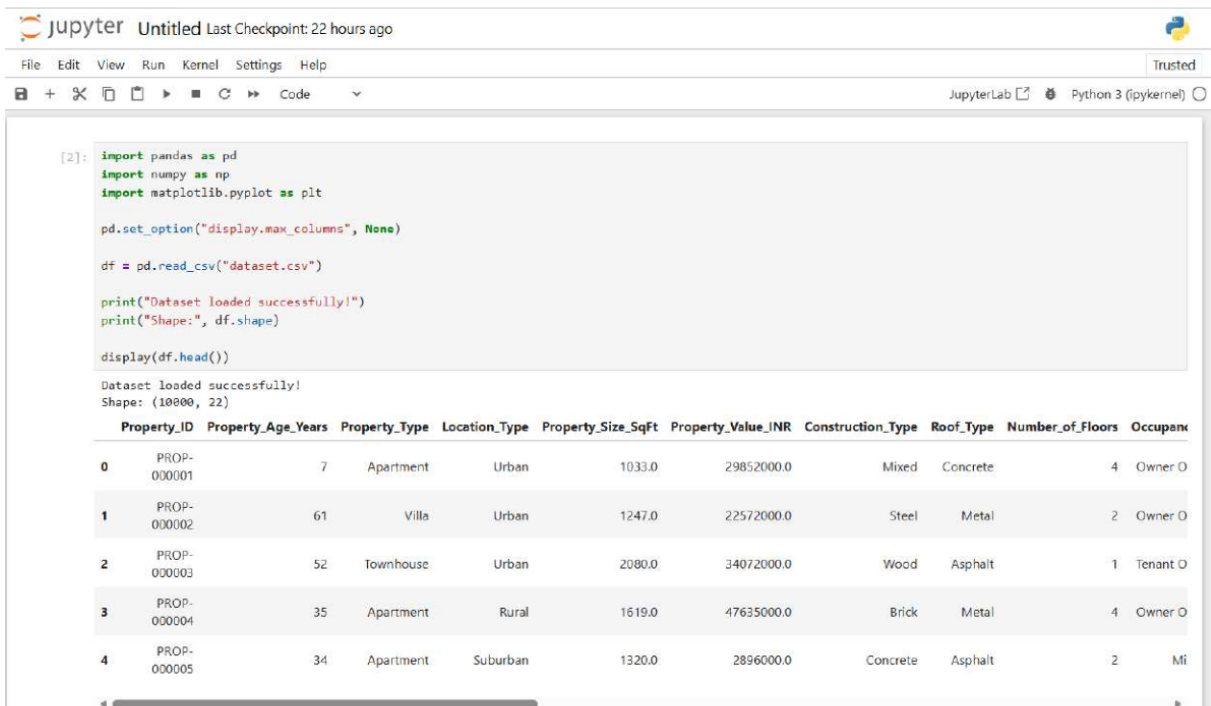


Figure 1: Initializing the environment, loading the dataset, and validating the 10,000 x 22 dimension.

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2.3 Schema and Data Type Validation

A critical inspection of the DataFrame schema via the `df.info()` method confirmed the initial structural integrity of the data. The output displays the precise memory usage (approximately 1.7 MB) and a breakdown of data types: 9 categorical/string variables, 8 floating-point variables (mostly financial metrics), and 5 integer variables (discrete counts like `Number_of_Floors` or `Age`). This explicit typing helps in mapping out our encoding strategies for Day 2.

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```
[3]:  
  
# Dataset information  
df.info()  
  
<class 'pandas.DataFrame'>  
RangeIndex: 10000 entries, 0 to 9999  
Data columns (total 22 columns):  
#   Column                                Non-Null Count  Dtype  
---  -  
0   Property_ID                           10000 non-null  str  
1   Property_Age_Years                     10000 non-null  int64  
2   Property_Type                           10000 non-null  str  
3   Location_Type                           10000 non-null  str  
4   Property_Size_SqFt                     10000 non-null  float64  
5   Property_Value_INR                     10000 non-null  float64  
6   Construction_Type                       10000 non-null  str  
7   Roof_Type                               9900 non-null   str  
8   Number_of_Floors                       10000 non-null  int64  
9   Occupancy_Type                         10000 non-null  str  
10  Security_Level                          8935 non-null   str  
11  Natural_Disaster_Risk                   10000 non-null  str  
12  Crime_Risk                              10000 non-null  str  
13  Previous_Claims_Count                   10000 non-null  int64  
14  Previous_Claim_Amount_INR               9900 non-null   float64  
15  Coverage_Amount_INR                     10000 non-null  float64  
16  Deductible_INR                           10000 non-null  float64  
17  Policy_Term_Years                       10000 non-null  int64  
18  Customer_Age                             10000 non-null  int64  
19  Customer_Income_INR                     10000 non-null  float64  
20  Credit_Score                             9900 non-null   float64  
21  Total_Annual_Premium_INR                10000 non-null  float64  
dtypes: float64(8), int64(5), str(9)  
memory usage: 1.7 MB  
  
[ ]:
```

Figure 2: Schema validation demonstrating column data types and non-null counts.

3. Data Quality & Integrity Assessment

3.1 Duplicate Record Analysis

To ensure the model does not suffer from sample bias or over-representation of specific properties, a thorough duplication check was performed. The execution of `df.duplicated().sum()` confirmed exactly 0 duplicate rows. This indicates a high degree of fidelity in the data extraction pipeline.

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[6]:

```
# Duplicate rows  
print("Duplicates:", df.duplicated().sum())
```

Duplicates: 0

[]:

|

Figure 3: Programmatic validation confirming zero duplicate observations.

3.2 Comprehensive Missing Value Profiling

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Real-world datasets are inherently messy. Identifying missing values early prevents catastrophic errors during matrix operations in machine learning algorithms. Our analysis exposed missing data across four key features:

- **Roof_Type** (100 Missing): A critical structural feature. Missing values may be due to unclassified modern roof designs or data entry oversight.
- **Security_Level** (1,065 Missing): Over 10% of this column is null. This significant gap suggests that either security data is optional for certain property types (e.g., rural areas) or it represents properties with 'No Security System', which we may need to encode as a distinct category rather than just a missing value.
- **Previous_Claim_Amount_INR** (100 Missing): This likely corresponds directly to customers who have 0 previous claims. We must investigate whether these nulls can be safely imputed with '0'.
- **Credit_Score** (100 Missing): Financial data is often withheld or unavailable. Mean or median imputation will be required to preserve the remaining financial profile of these clients.

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[5]:

```
# Missing values
print(df.isnull().sum())
```

```
Property_ID                0
Property_Age_Years         0
Property_Type              0
Location_Type              0
Property_Size_SqFt         0
Property_Value_INR         0
Construction_Type          0
Roof_Type                  100
Number_of_Floors           0
Occupancy_Type             0
Security_Level            1065
Natural_Disaster_Risk      0
Crime_Risk                 0
Previous_Claims_Count      0
Previous_Claim_Amount_INR  100
Coverage_Amount_INR        0
Deductible_INR             0
Policy_Term_Years          0
Customer_Age               0
Customer_Income_INR        0
Credit_Score              100
Total_Annual_Premium_INR   0
dtype: int64
```

[]:

|

Figure 4: Missing value counts highlighting the sparsity in Security_Level and other features.

4. Statistical Profiling & Feature Categorization

4.1 Analyzing Numerical Features

Using `df.describe()`, we generated a five-number summary for all numerical features. Several immediate business and statistical insights were drawn:

- **Property Age:** Ranges from 0 (brand new builds) to 79 years, indicating a highly diverse portfolio of insured properties. The mean age is ~39 years.
- **Property Value (INR):** The valuation shows extreme variance, stretching up to 100,000,000 INR (10 Crore). This right-skewed variance indicates the presence of luxury properties, which may act as statistical outliers impacting the regression weightings.
- **Customer Demographics:** The customer age ranges appropriately, while Customer Income shows massive variance, indicating our model must be resilient to differing economic profiles.

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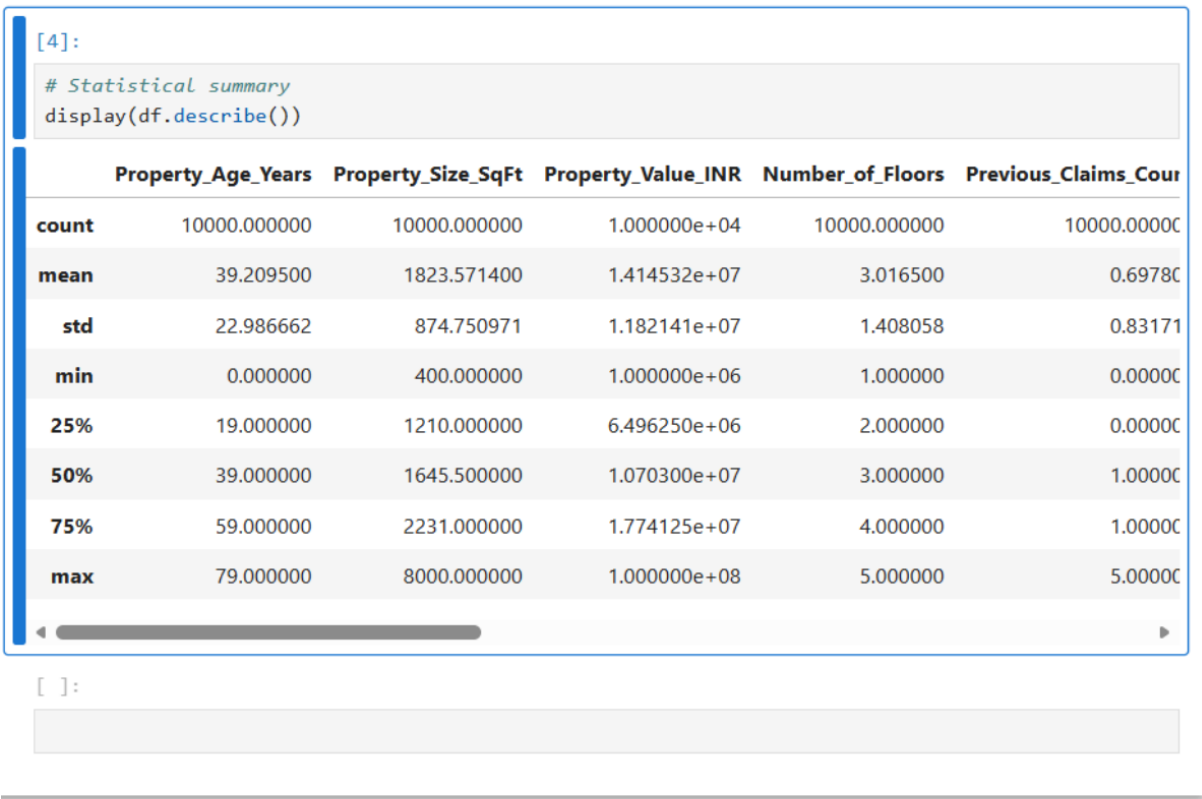


Figure 5: Statistical dispersion, central tendencies, and quartile breakdowns for numerical variables.

4.2 Feature Type Isolation

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To streamline the Day 2 preprocessing pipeline, we programmatically isolated columns into continuous (numerical) and discrete (categorical) lists using ``select_dtypes()``. This allows us to cleanly map our Scaling (Standard/MinMax) and Encoding (One-Hot/Label) pipelines.

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```
[11]: # Numerical columns
print("Numerical columns:")
print(df.select_dtypes(include=["int64", "float64"]).columns.tolist())

Numerical columns:
['Property_Age_Years', 'Property_Size_SqFt', 'Property_Value_INR', 'Number_of_Floors', 'Previous_Claims_Count', 'Previous_Claim_Amount_INR', 'Coverage_Amount_INR', 'Deductible_INR', 'Policy_Term_Years', 'Customer_Age', 'Customer_Income_INR', 'Credit_Score', 'Total_Annual_Premium_INR']
```

Figure 6: Isolating numerical columns for mathematical transformations.

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```

[14]: categorical_columns = df.select_dtypes(include=["str"]).columns.tolist()
      print("Categorical columns:")
      print(categorical_columns)

Categorical columns:
['Property_ID', 'Property_Type', 'Location_Type', 'Construction_Type', 'Roof_Type', 'Occupancy_Type', 'Security_Level', 'Natural_Disaster_Risk', 'Crime_Risk']
```

Figure 7: Isolating categorical string columns for structural encoding strategies.

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5. Target Variable Deep Dive: Annual Premium

5.1 Statistical Breakdown of the Target

The ultimate goal of this project is to predict 'Total_Annual_Premium_INR'. A dedicated statistical analysis of this single vector is vital. The metrics reveal a mean premium of 66,980 INR and a median (50th percentile) of 49,400 INR. When the mean is significantly higher than the median, it mathematically proves a right-skewed (positive skew) distribution. The massive maximum value of 701,600 INR further confirms a 'long tail' of highly expensive premiums, likely tied to the luxury properties identified earlier.

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```
[15]: target = "Total_Annual_Premium_INR"
      display(df[target].describe())
```

count	10000.000000
mean	66980.980000
std	60368.311833
min	5000.000000
25%	28900.000000
50%	45400.000000
75%	83100.000000
max	701600.000000

Name: Total_Annual_Premium_INR, dtype: float64

Figure 8: Focused descriptive statistics for the 'Total_Annual_Premium_INR' target variable.

5.2 Visualizing the Distribution

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To confirm the statistical hypothesis, a Matplotlib histogram with 30 bins was plotted. The visual output unequivocally demonstrates the right-skewed nature of the data. The vast majority of insurance premiums fall into the lower brackets (between 5,000 and 100,000 INR), with frequency sharply dropping off as the premium increases.

From a Machine Learning perspective, predicting on highly skewed target variables can cause regression models (especially linear variants) to perform poorly, as they optimize for Mean Squared Error (MSE) and become disproportionately punished by these outliers. A logarithmic transformation (Log_{10}) of this target variable will likely be necessary in Phase 2 to normalize the curve.

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```
[16]: plt.figure(figsize=(10, 6))  
  
plt.hist(df[target], bins=30)  
  
plt.title("Distribution of Annual Insurance Premium")  
plt.xlabel("Annual Premium (INR)")  
plt.ylabel("Frequency")  
  
plt.show()
```

Figure 9: Histogram visualizing the heavily right-skewed probability distribution of the premiums.

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6. Conclusion & Comprehensive Day 2 Action Plan

Day 1 successfully established the integrity, structure, and statistical reality of the insurance dataset. We have identified critical data nuances, such as missing security information and extreme right-skewness in property valuations and target premiums. To transition from raw data to a machine-learning-ready state, Day 2 will execute the following rigorous pipeline:

- 1. Advanced Imputation: We will impute 'Credit_Score' via median values to resist outliers. For 'Security_Level', the 1,065 missing records will be converted into a new distinct categorical class (e.g., 'Unknown_Security') rather than purely imputed, as the 'missingness' itself might hold predictive power.
- 2. Target Transformation: We will apply a $\text{np.log1p}()$ transformation to 'Total_Annual_Premium_INR' to force it into a Gaussian (Normal) distribution, greatly improving the predictive accuracy of linear algorithms.
- 3. Multivariate Outlier Detection: We will utilize Boxplots and IQR (Interquartile Range) clipping to cap extreme values in 'Property_Value_INR' without losing valuable data.
- 4. Feature Engineering & Encoding: Applying One-Hot Encoding to non-ordinal categories (e.g., 'Roof_Type') and Ordinal Encoding to ranked features (e.g., 'Security_Level' if applicable).
- 5. Correlation & Feature Selection: Plotting a Seaborn Heatmap to identify multicollinearity between independent variables and dropping redundant features before model training.

--- End of Day 1 Comprehensive Report ---